

National Health and Nutrition Examination Survey

2017-2018 Data Documentation, Codebook, and Frequencies

Plasma Fasting Glucose (GLU_J)

Data File: GLU_J.xpt

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Component Description

In 2015, diabetes was the seventh leading cause of death in the United States. More than 30 million Americans are living with diabetes, and 86 million are living with prediabetes, which is a serious health condition that increases a person's risk of type-2 diabetes and other chronic diseases. The prevalence of diabetes and overweight (one of the major risk factors for diabetes) continues to increase. Substantial new efforts to prevent or control diabetes have begun, including the Diabetes Prevention Trial and the National Diabetes Education Program.

Diabetes testing provides population data to: 1) determine a national estimate of diabetes prevalence (diagnosed and undiagnosed); 2) identify the risk factors; 3) permit a national cohort to be established for follow-up studies of this condition; and 4) provide critical information to clinicians and public health officials for the development of preventive care and community-based interventions.

Eligible Sample

Participants aged 12 years and older, who were examined in the morning session, were eligible.

Description of Laboratory Methodology

Glucose

In this enzymatic method glucose is converted to glucose-6-phosphate (G-6-P) by hexokinase in the presence of ATP, a phosphate donor. Glucose-6-phosphate dehydrogenase then converts the G-6-P to gluconate-6-P in the presence of NADP+. As the NADP+ is reduced to NADPH during this reaction, the resulting increase in absorbance at 340 nm (secondary wavelength = 700 nm) is measured. This is an endpoint reaction that is specific for glucose.

Refer to the Laboratory Method Files section for a detailed description of the laboratory methods used.

There were no changes to the lab site or lab method for this component in the NHANES 2017-2018 cycle. There were no changes to the lab equipment since 2015-2016. The Cobas C311 analyzer was used in 2017-2018 and 2015-2016. Refer to analytic note for changes to equipment in earlier survey cycles, which may impact analysis.

Laboratory Method Files

[Plasma Fasting Glucose Laboratory Procedure Manual](#) (August 2020)

Laboratory Quality Assurance and Monitoring

Plasma specimens are processed, stored, and shipped to the University of Missouri-Columbia, Columbia, MO for analysis.

Detailed instructions on specimen collection and processing are discussed in the [NHANES Laboratory Procedures Manual \(LPM\)](#). Vials are stored under appropriate frozen (-30°C) conditions until they are shipped to University of Missouri-Columbia for testing.

The NHANES quality assurance and quality control (QA/QC) protocols meet the 1988 Clinical Laboratory Improvement Act mandates. Detailed QA/QC instructions are discussed in the [NHANES LPM](#).

Mobile Examination Centers (MECs)

Laboratory team performance is monitored using several techniques. NCHS and contract consultants use a structured competency assessment evaluation during visits to evaluate both the quality of the laboratory work and the quality control procedures. Each laboratory staff member is observed for equipment operation, specimen collection and preparation; testing procedures and constructive feedback are given to each staff member. Formal retraining sessions are conducted annually to ensure that required skill levels were maintained.

Analytical Laboratories

NHANES uses several methods to monitor the quality of the analyses performed by the contract laboratories. In the MEC, these methods include performing blind split samples collected on "dry run" sessions. In addition, contract laboratories randomly perform repeat testing on 2% of all specimens.

NCHS developed and distributed a QC protocol for all CDC and contract laboratories, which outlined the use of Westgard rules (Westgard et al, 1981) when running NHANES specimens. Progress reports containing any problems encountered during shipping or receipt of specimens, summary statistics for each control pool, QC graphs, instrument calibration, reagents, and any special considerations are submitted to NCHS quarterly. The reports are reviewed for trends or shifts in the data. The laboratories are required to explain any identified areas of concern.

Data Processing and Editing

The data were reviewed. Incomplete data or improbable values were sent to the performing laboratory for confirmation.

One variable was created in this data file. The variable was created using the following formulas:

LBXGLU and LBDGLUSI:

The fasting glucose value in mg/dL (LBXGLU) was converted to mmol/L (LBDGLUSI) by multiplying by 0.05551 (rounded to 3 decimals).

Analytic Notes

Refer to the [2017 - 2018 Laboratory Data Overview](#) for general information on NHANES laboratory data.

There are over 800 laboratory tests performed on NHANES participants. However, not all participants provided biospecimens or enough volume for all the tests to be performed. The specimen availability can also vary by age or other population characteristics. For example, in 2017-2018, approximately 80% of children aged 1-17 years who were examined in the MEC provided a blood specimen through phlebotomy, while 95% of examined adults age 18 and older provided a blood specimen. Analysts should evaluate the extent of missing data in the

dataset related to the outcome of interest as well as any predictor variables used in the analyses to determine whether additional re-weighting for item non-response is necessary.

Please refer to the NHANES [Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Subsample Weights

Glucose was measured in a fasting subsample of participants 12 years and older. Specific sample weights (WTSAF2YR) for this subsample are included in this data file and should be used when analyzing these data. Participants included in the fasting subsample but did not provide a blood specimen (n=145) have an assigned sample weight value of "0" in their records. In addition, participants whom provided blood specimen but did not meet the 8 to less than 24 hour fasting criteria (n=180) have the sample weight value assigned as "0" (WTSAF2YR =0) as well. Glucose measurements for these 180 participants are included in the dataset. However, to include these data in the analysis, a reweighting would be required.

Demographic and Other Related Variables

The analysis of NHANES laboratory data must be conducted using the appropriate survey design and demographic variables. The [NHANES 2017-2018 Demographics File](#) contains demographic data, health indicators, and other related information collected during household interviews as well as the sample design variables. The recommended procedure for variance estimation requires use of stratum and PSU variables (SDMVSTRA and SDMVPSU, respectively) in the demographic data file.

In the 2017-2018 NHANES cycle, the variables PHAFSTHR (total length of "food fast", hours) and PHAFSTMN (Total length of "food fast", minutes) will not be reported in this file. PHAFSTHR and PHAFSTMN can be found in the [Fasting Questionnaire File](#) (FASTQX_J). The fasting questionnaire data file also includes additional auxiliary information such as fasting status, and the time of venipuncture.

This laboratory data file can be linked to the other NHANES data files using the unique survey participant identifier (i.e., SEQN).

Detection Limits

The detection limits were constant for all of the analytes in the data set. Two variables are provided for each of these analytes. The variable name ending "LC" (ex., LBDGLULC) indicates whether the result was below the limit of detection: the value "0" means that the result was at or above the limit of detection, "1" indicates that the result was below the limit of detection. The other variable prefixed LBX (ex., LBXGLU) provides the analytic result for that analyte. For analytes with analytic results below the lower limit of detection (ex., LBDGLULC=1), an imputed fill value was placed in the analyte results field. This value is the lower limit of detection divided by the square root of 2 (LLOD/sqrt[2]).

The lower limit of detection (LLOD, in mg/dL) for fasting glucose:

Variable Name	SAS LABEL	LLOD
LBXGLU	Fasting glucose	2.0

Comparability with previous cycles

There were equipment changes during the 2015-2016, 2007-2008, and 2005-2006 survey cycles. For analysis involving 2015-2016 data and data collected prior to 2007-2008 cycle,

please refer to the documentation accompanying those data for additional information on any needed adjustments to the data prior to analysis. If 2017-2018 data are compared or combined with the 2015-2016 data, no adjustments are needed for equipment changes. As mentioned in the 2015-2016 documentation, most of the 2015-2016 samples were measured using the Cobas C311, which is the same instrument used in 2017-2018. Some of the samples were collected using the Cobas C501. Results for these specimens have already been adjusted using the forward regression equation for comparability. Please see the 2015-2016 fasting plasma glucose documentation for more information about using backwards or forwards regression models to account for equipment changes between cycles (include link to documentation here).

References

- Westgard J.O., Barry P.L., Hunt M.R., Groth T. A multi-rule Shewhart chart for quality control in clinical chemistry. Clin Chem (1981) 27:493-501.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number
Target:	Both males and females 12 YEARS - 150 YEARS

WTSAF2YR - Fasting Subsample 2 Year MEC Weight

Variable Name: WTSAF2YR

SAS Label: Fasting Subsample 2 Year MEC Weight

English Text: Fasting Subsample 2 Year MEC Weight

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
9133.518063 to 944153.24975	Range of Values	2711	2711	
0	No Lab Result or Not Fasting for 8 to <24 hours	325	3036	
.	Missing	0	3036	

LBXGLU - Fasting Glucose (mg/dL)**Variable Name:** LBXGLU**SAS Label:** Fasting Glucose (mg/dL)**English Text:** Fasting Glucose (mg/dL)**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
47 to 451	Range of Values	2891	2891	
.	Missing	145	3036	

LBDGLUSI - Fasting Glucose (mmol/L)

Variable Name: LBDGLUSI

SAS Label: Fasting Glucose (mmol/L)

English Text: Fasting Glucose (mmol/L)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
2.61 to 25	Range of Values	2891	2891	
.	Missing	145	3036	